Methods for Handling Multiple Outcomes in Health Data

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1 Introduction

1.1 Background

Clinical research frequently involves more than one outcome of interest. In oncology and chronic disease studies, patients may experience several clinically important events during follow-up, including disease progression, recurrence, metastasis, hospitalisation, and death. These outcomes are often analysed separately using conventional methods, but separate analyses may not fully reflect the patient journey because events can be clinically and statistically related (1–3).

Time-to-event outcomes are especially important in cancer research because not all patients experience the event during the observed follow-up period. Patients who do not experience the event are censored, meaning that their event time is only partially observed. Survival methods such as Kaplan–Meier estimation, log-rank tests, and Cox proportional hazards regression are therefore widely used in clinical trials (4–7). However, these methods are usually applied to one endpoint at a time.

1.2 Multiple Outcomes in Clinical Research

Multiple outcomes matter because they may represent different levels of clinical importance. For example, death is usually more serious than disease progression, while progression may still be important because it can reduce quality of life, lead to treatment switching, and increase the future risk of death. A statistical method for multiple outcomes should ideally account for timing, censoring, clinical hierarchy, and the dependency between outcomes (8–10).

Composite endpoints are often used to combine several events into a single endpoint. A participant is considered to have the composite event if any component occurs. This can increase the number of events and statistical power, but it may also be misleading when components differ substantially in clinical importance (2,3,11). For example, a composite endpoint combining progression and death may be driven mainly by progression events, even though death is the more serious outcome.

1.3 Challenges in Analysing Multiple Outcomes

Analysing multiple outcomes creates several methodological challenges. First, repeated testing of separate endpoints can increase the probability of false positive findings. Second, outcomes may be dependent; for example, disease progression may alter the subsequent risk of death. Third, treatment effects may differ across endpoints or change over time. Fourth, proportional hazards assumptions may not always hold in survival models, especially when survival curves cross (12–14).

Advanced methods have been developed to address these challenges. The win ratio compares patients using a pre-specified hierarchy of outcomes, giving priority to the most clinically important event (8). Net-benefit approaches provide decision-focused summaries of clinical advantage (15,16). Multi-state models describe how patients move between disease states over time and can estimate transition-specific hazards and probabilities (14,17–19).

1.4 SECOMBIT Trial Overview

This project used data from the SECOMBIT trial, an open-label, multicentre, randomised phase II clinical trial involving patients with metastatic BRAF V600-mutated melanoma. The trial evaluated treatment sequencing strategies involving targeted therapy with encorafenib plus binimetinib and immunotherapy with ipilimumab plus nivolumab (20,21).

The trial included three treatment arms. Arm A received encorafenib plus binimetinib followed by ipilimumab plus nivolumab after disease progression. Arm B received ipilimumab plus nivolumab followed by encorafenib plus binimetinib after disease progression. Arm C used a planned "sandwich" strategy: patients received encorafenib plus binimetinib for eight weeks and then switched to ipilimumab plus nivolumab until progression, followed by encorafenib plus binimetinib after progression (20,21).

1.5 Dataset Description

The dataset used in this project was obtained from an open-source version of the SECOMBIT clinical trial data made available for research and educational analysis. The dataset contained approximately 209 patients allocated to Arms A, B, and C. The key outcomes were overall survival (OS) and progression-free survival (PFS). Overall survival was defined as the time from study entry to death, while progression-free survival was defined as the time from study entry to disease progression or death. The main covariates available for analysis were treatment arm, number of metastatic sites, lactate dehydrogenase (LDH) level, tumour mutational burden (TMB), and JAK mutation status.

The dataset was suitable for this project because it contained multiple clinically important time-to-event outcomes that are related but not equivalent. It also allowed direct comparison between traditional survival methods and more advanced approaches such as win-ratio analysis, net-benefit analysis, and multi-state modelling.

1.6 Aim and Objectives

The aim of this project was to compare statistical strategies for handling multiple outcomes in health data, using the SECOMBIT dataset as an applied case study.

The objectives were:

1. To analyse OS and PFS separately using logistic regression and survival analysis.
2. To compare individual endpoint analysis with composite endpoint analysis.
3. To evaluate treatment-arm differences using Cox proportional hazards models.
4. To apply hierarchical and decision-focused approaches using win-ratio and net-benefit methods.
5. To model disease pathways using a multi-state model.
6. To compare the strengths, limitations, and clinical interpretability of each method.

2 Methods

2.1 Data Preparation

The SECOMBIT dataset was imported into R and cleaned before analysis. Variables were recoded to create standard event-time and event-status variables for OS, PFS, and the composite endpoint. Treatment arm and clinical covariates were coded as categorical variables. A composite endpoint was created by defining an event when either progression or death occurred.

2.2 Missing Data Handling

Missing data were assessed before fitting adjusted models. Missing values were present mainly in the JAK mutation variable. No formal imputation was performed. Therefore, regression models requiring all covariates used complete-case analysis. This means that participants with missing values in any covariate included in the model were excluded from that specific analysis.

Complete-case analysis is simple and transparent, but it can reduce statistical power and may introduce bias if data are not missing completely at random (22,23). This limitation is especially relevant for multivariable

Cox and logistic regression models, where the inclusion of JAK mutation status reduced the effective sample size.

Table 1: Missing values by variable

Variable	Missing values
id	0
Arm	0
Sites	2
LDH	5
TMB	126
JAK	126
pfs_time	3
pfs_status	3
os_time	3
os_status	3
comp_time	3
comp_status	3
binary_os	3
binary_pfs	3
binary_comp	3

2.3 Statistical Methods

2.3.1 Logistic Regression

Logistic regression was used as a preliminary event-based analysis for OS, PFS, and the composite endpoint. For a binary event indicator, the model can be written as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip},$$

where p_i is the probability that patient i experiences the event and $X_{i1}, \ldots, X_{ip}$ are covariates. Odds ratios greater than 1 indicate higher odds of the event, while odds ratios less than 1 indicate lower odds. Logistic regression does not account for time-to-event information or censoring, so it was not treated as the primary method for survival outcomes (24).

2.3.2 Kaplan–Meier Estimation

Kaplan–Meier estimation was used to estimate survival probabilities over time (4). The Kaplan–Meier estimator is:

$$\hat{S}(t) = \prod_{t_i \le t}\left(1 - \frac{d_i}{n_i}\right),$$

where d_i is the number of events at time t_i and n_i is the number of individuals at risk just before t_i. Kaplan–Meier curves were generated for OS, PFS, and the composite endpoint by treatment arm.

2.3.3 Log-Rank Test

The log-rank test was used to compare survival curves across treatment arms (5). It tests the null hypothesis that there is no difference in survival experience between groups. The test is sensitive to differences in hazards over time, but it does not directly estimate the size or direction of the treatment effect.

2.3.4 Cox Proportional Hazards Model

Cox proportional hazards models were used to estimate adjusted treatment effects for OS, PFS, and the composite endpoint (6). The model is:

$$h(t|X) = h_0(t)\exp(\beta X),$$

where $h(t|X)$ is the hazard at time t for covariate vector X, $h_0(t)$ is the baseline hazard, and β represents the log hazard ratio. Hazard ratios greater than 1 indicate increased instantaneous risk, while hazard ratios less than 1 indicate reduced instantaneous risk.

The key assumption is proportional hazards, meaning that hazard ratios remain constant over follow-up time. This assumption was assessed using Schoenfeld residuals (12,25).

2.3.5 Composite Endpoint Analysis

The composite endpoint was defined as the first occurrence of progression or death. Composite endpoints can increase the number of events and simplify interpretation, but they may be problematic when the components differ in clinical importance (2,3,11). In this project, the composite endpoint was interpreted carefully because progression and death are not clinically equivalent.

2.3.6 Win Ratio

The win ratio is a hierarchical method for comparing treatment groups using multiple outcomes (8). Patients are compared pairwise. The most clinically important outcome is evaluated first; if a pair is tied, the next outcome is considered. In this project, OS was prioritised first and PFS second.

The win ratio is defined as:

$$WR = \frac{\text{Number of wins}}{\text{Number of losses}}.$$

A win ratio greater than 1 favours the treatment group, while a value less than 1 favours the comparator group. This method is clinically meaningful because it respects the hierarchy of outcomes.

2.3.7 Net Benefit

Net benefit was estimated using generalised pairwise comparisons through the `BuyseTest` package. Net benefit can be expressed as:

$$NB = P(\text{Win}) - P(\text{Loss}),$$

where $P(\text{Win})$ is the probability that a patient in one group has a better outcome than a comparable patient in another group, and $P(\text{Loss})$ is the probability of a worse outcome (16,26). Positive net benefit indicates an overall clinical advantage.

2.3.8 Multi-State Models

Multi-state models were used to model transitions between disease states over time (14,17,18). The states were stable disease, progression, and death. The transition intensity from state r to state s is:

$$q_{rs} = \lim_{\Delta t \to 0} \frac{P(X(t + \Delta t) = s | X(t) = r)}{\Delta t}.$$

The model estimated transition-specific hazards and state occupation probabilities. A Markov assumption was used, meaning that future transitions depend on the current state and covariates rather than the full previous history.

2.4 State-Transition Structure

The intended clinical pathway can be represented as follows:

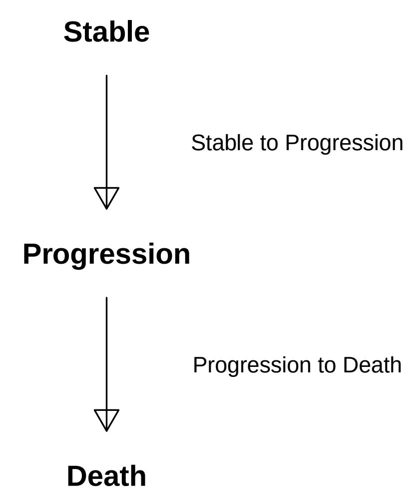

Figure 1: State-transition diagram for the multi-state model

Although a direct transition from stable disease to death can occur in some oncology datasets, no such transition was observed in this dataset. Therefore, the fitted model allowed transitions from stable disease to progression and from progression to death.

2.5 Software Implementation

All analyses were conducted in R. The main packages used were `survival` for survival modelling, `survminer` for Kaplan–Meier plots, `msm` for multi-state modelling, `WinRatio` for win-ratio analysis, `BuyseTest` for net-benefit analysis, `tableone` for descriptive baseline tables, `broom` for model summaries, and `knitr`/`kableExtra` for table formatting (17,27–30).

3 Results

3.1 Descriptive Summary

Table 2: Numeric summary of OS and PFS in months

Statistic	OS	PFS
mean	32.39	28.81
sd	17.19	18.06
median	35.90	29.80
min	1.13	1.10
max	65.50	65.50

3.2 Baseline Characteristics

Only variables measured at or before treatment initiation were included in the baseline table. Outcome variables such as OS status and PFS status were excluded because they occur after treatment begins.

Table 3: Baseline characteristics by treatment arm

	level	A	B	C	p	test
n		69	71	69		
Sites (%)	1 - 2	43 (62.3)	41 (58.6)	42 (61.8)	0.887	
	>= 3	26 (37.7)	29 (41.4)	26 (38.2)		
LDH (%)	Normal	41 (61.2)	41 (59.4)	48 (70.6)	0.346	
	Elevated	26 (38.8)	28 (40.6)	20 (29.4)		
TMB (%)	< 10	20 (71.4)	17 (68.0)	18 (60.0)	0.639	
	>= 10	8 (28.6)	8 (32.0)	12 (40.0)		
JAK (%)	Wild Type	24 (82.8)	17 (70.8)	16 (53.3)	0.050	
	Mutated	5 (17.2)	7 (29.2)	14 (46.7)		

3.3 Event Counts by Treatment Arm

Table 4: Event counts by treatment arm

Arm	n	OS_Events	PFS_Events	Composite_Events
A	69	35	49	49
B	71	23	30	30
C	69	27	32	32

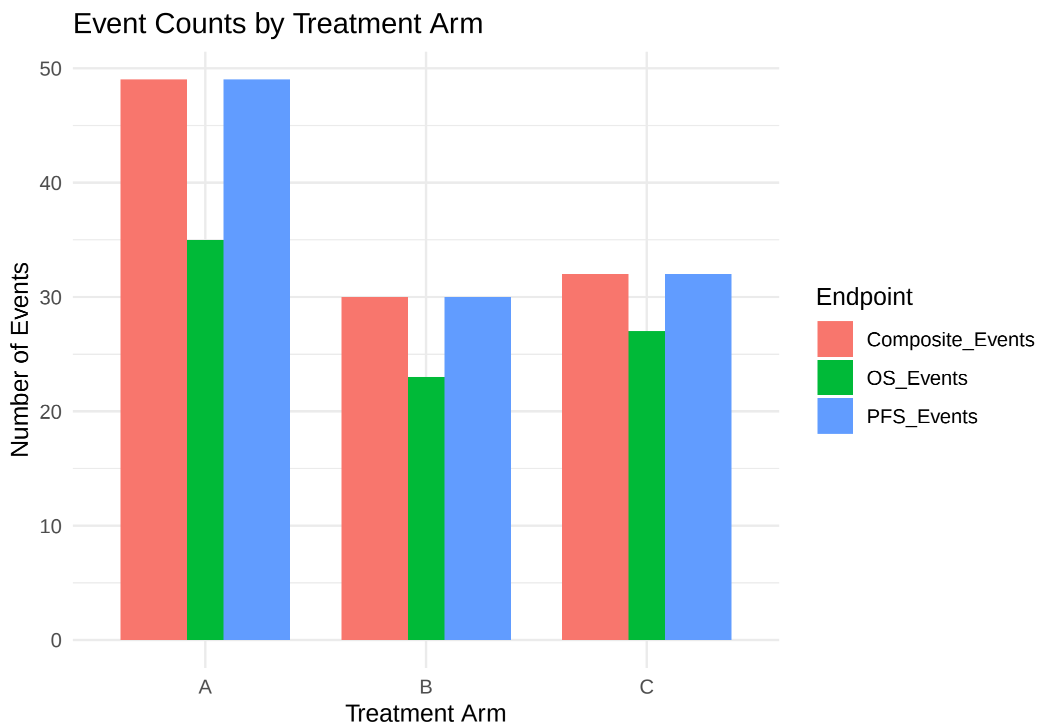

Figure 2: OS, PFS, and composite event counts by treatment arm

3.4 Logistic Regression Results

Table 5: Logistic regression for OS event

Variable	OR (95% CI)	p-value
(Intercept)	0.64 (0.25–1.61)	0.3490
ArmB	0.70 (0.20–2.34)	0.5700
ArmC	0.83 (0.25–2.70)	0.7500
Sites>= 3	2.45 (0.88–7.13)	0.0908
LDHElevated	0.79 (0.26–2.27)	0.6650
TMB>= 10	0.71 (0.23–2.03)	0.5230
JAKMutated	0.52 (0.15–1.64)	0.2730

Table 6: Logistic regression for PFS event

Variable	OR (95% CI)	p-value
(Intercept)	1.33 (0.53–3.42)	0.5450
ArmB	0.43 (0.12–1.44)	0.1800
ArmC	1.08 (0.34–3.59)	0.8940
Sites>= 3	1.83 (0.65–5.37)	0.2590
LDHElevated	0.90 (0.30–2.62)	0.8410
TMB>= 10	0.72 (0.25–2.07)	0.5460
JAKMutated	0.26 (0.07–0.80)	0.0247

Table 7: Logistic regression for composite event

Variable	OR (95% CI)	p-value
(Intercept)	1.33 (0.53–3.42)	0.5450
ArmB	0.43 (0.12–1.44)	0.1800
ArmC	1.08 (0.34–3.59)	0.8940
Sites>= 3	1.83 (0.65–5.37)	0.2590
LDHElevated	0.90 (0.30–2.62)	0.8410
TMB>= 10	0.72 (0.25–2.07)	0.5460
JAKMutated	0.26 (0.07–0.80)	0.0247

Logistic regression was useful as a preliminary event-based analysis, but it ignored event timing and censoring. Therefore, survival models were more appropriate for the main OS and PFS analyses.

3.5 Kaplan–Meier Survival Curves and Log-Rank Tests

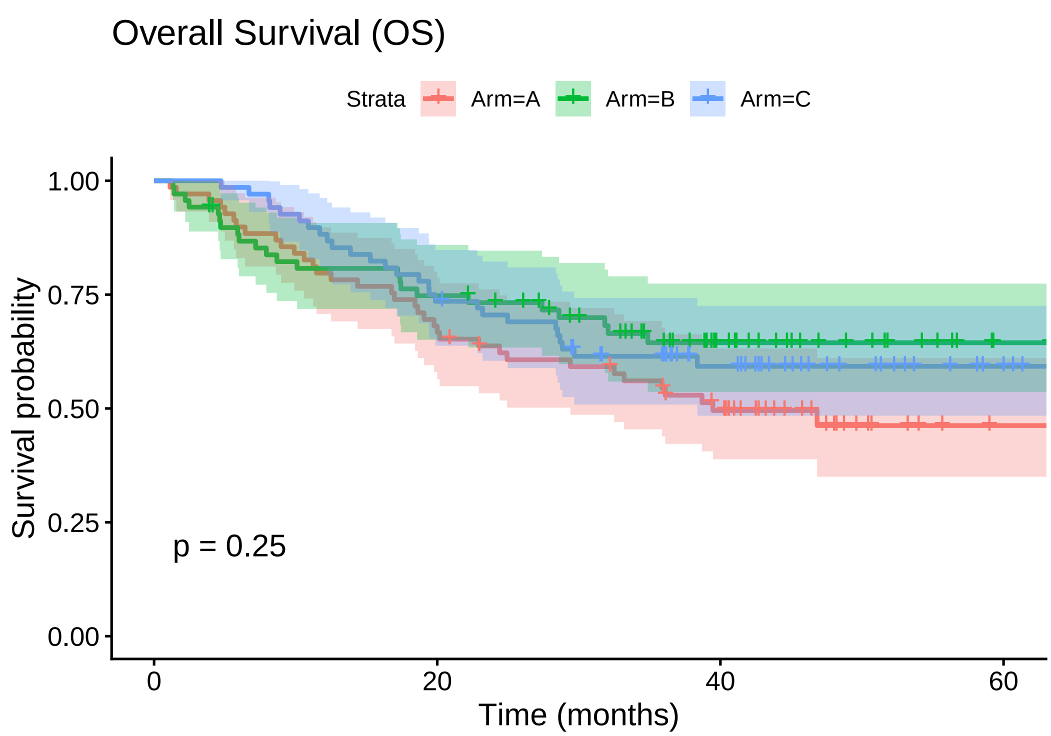

Figure 3: Kaplan–Meier curve for overall survival by treatment arm

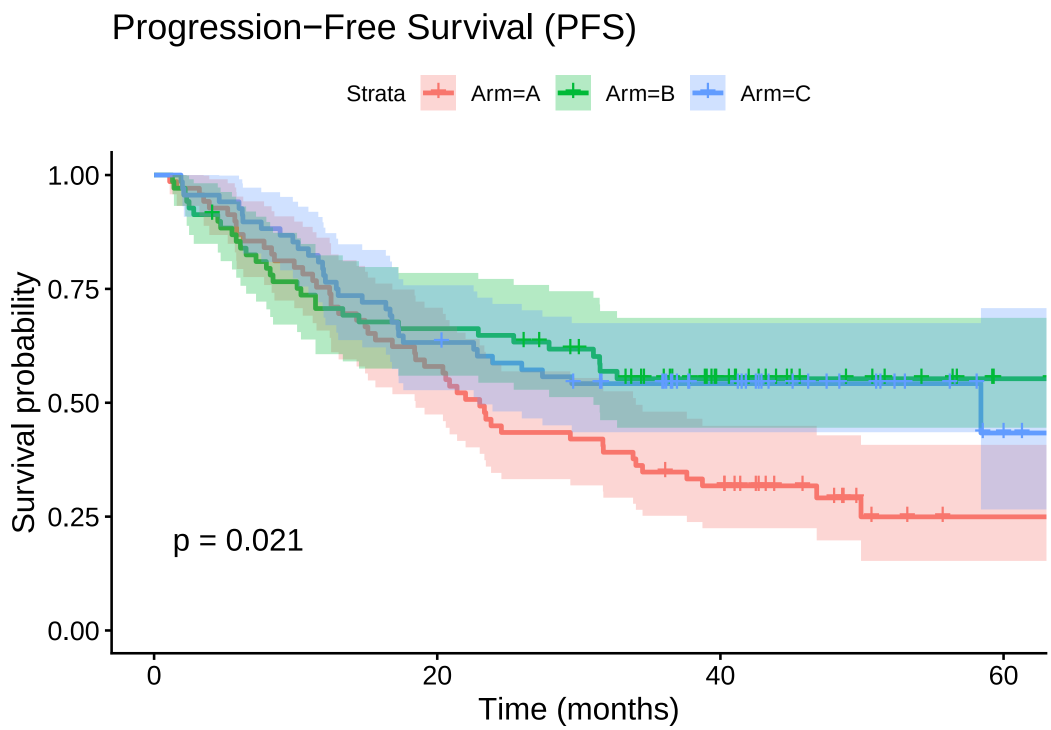

Figure 4: Kaplan–Meier curve for progression-free survival by treatment arm

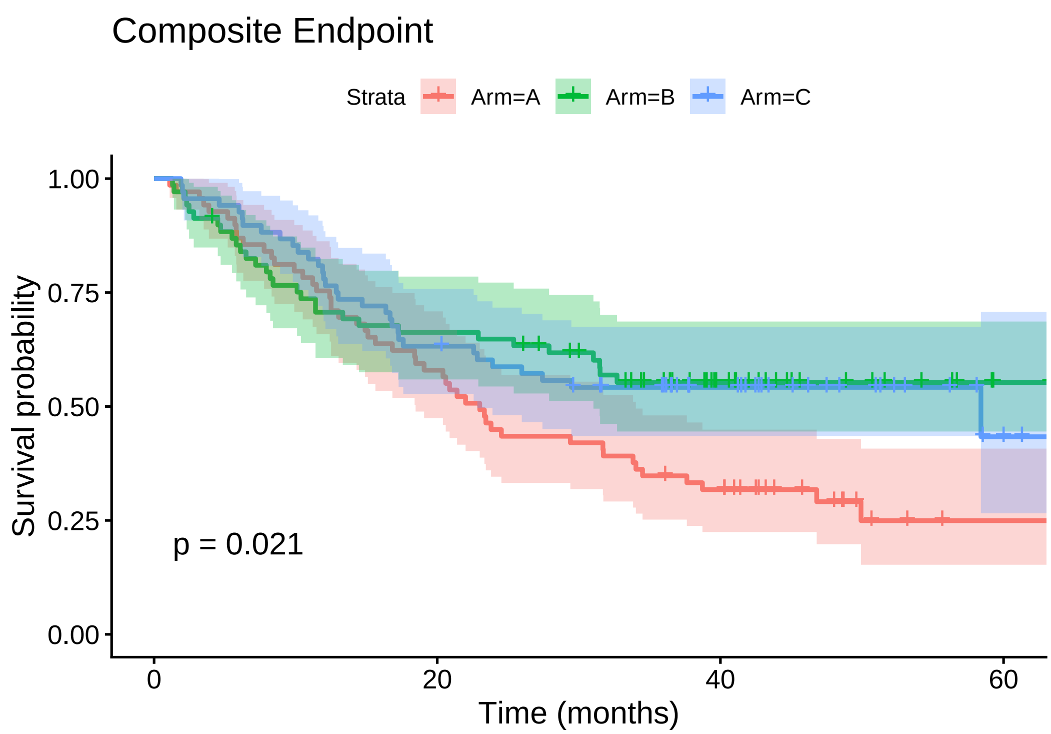

Figure 5: Kaplan–Meier curve for composite endpoint by treatment arm

Table 8: Log-rank test results

Endpoint	Chi.square	df	p.value
OS	2.80	2	0.2460
PFS	7.74	2	0.0208
Composite Endpoint	7.74	2	0.0209

The Kaplan–Meier curves for OS appeared relatively close, and the OS log-rank p-value was approximately 0.25, suggesting no statistically significant difference in overall survival across arms. The PFS curves separated more clearly, with a log-rank p-value near 0.02, indicating evidence that at least one arm differed in progression-free survival.

3.6 Cox Proportional Hazards Models

Table 9: Cox model for OS

Variable	HR (95% CI)	p-value
ArmB	0.98 (0.38–2.50)	0.9590
ArmC	0.83 (0.33–2.07)	0.6900
Sites>= 3	2.02 (0.93–4.42)	0.0764
LDHElevated	0.96 (0.43–2.15)	0.9130
TMB>= 10	0.72 (0.31–1.67)	0.4480
JAKMutated	0.63 (0.24–1.63)	0.3380

Table 10: Cox model for PFS

Variable	HR (95% CI)	p-value
ArmB	0.73 (0.30–1.77)	0.4870
ArmC	1.03 (0.48–2.22)	0.9470
Sites>= 3	1.60 (0.80–3.20)	0.1870
LDHElevated	1.14 (0.56–2.34)	0.7150
TMB>= 10	0.85 (0.40–1.78)	0.6660
JAKMutated	0.41 (0.16–1.03)	0.0571

Table 11: Cox model for composite endpoint

Variable	HR (95% CI)	p-value
ArmB	0.73 (0.30–1.77)	0.4870
ArmC	1.03 (0.48–2.22)	0.9470
Sites>= 3	1.60 (0.80–3.20)	0.1870
LDHElevated	1.14 (0.56–2.34)	0.7150
TMB>= 10	0.85 (0.40–1.78)	0.6660
JAKMutated	0.41 (0.16–1.03)	0.0571

In the Cox model for OS, the hazard ratios for Arm B and Arm C compared with Arm A were close to 1 with wide confidence intervals and non-significant p-values, suggesting no clear overall survival advantage in the adjusted model. For PFS, Arm B and JAK mutation showed protective trends, while greater metastatic-site burden and elevated LDH showed increased risk trends.

Table 12: Proportional hazards assumption tests

Endpoint	PH.global.p.value
OS	0.0581
PFS	0.0119
Composite Endpoint	0.0119

The proportional hazards test suggested a borderline violation for OS and a clearer violation for PFS. This indicates that treatment effects may vary over follow-up time, supporting the use of additional approaches such as multi-state modelling.

3.7 Win-Ratio Results

Table 13: Win-ratio comparison of treatment arms

Comparison	Win ratio (95% CI)	p-value
B vs A	1.49 (0.93–2.40)	0.0997
C vs A	1.59 (1.00–2.52)	0.0503
C vs B	1.06 (0.61–1.84)	0.8372

The win-ratio analysis favoured Arm B over Arm A, with a win ratio of 1.49 (95% CI 0.93–2.40; p = 0.0997). Arm C also favoured over Arm A, with a win ratio of 1.59 (95% CI 1.00–2.52; p = 0.0503). Arm C compared with Arm B was close to neutral, with a win ratio of 1.06 (95% CI 0.61–1.84; p = 0.8372). Overall, Arms B and C showed favourable hierarchical outcome trends compared with Arm A, but there was no clear superiority between Arms B and C.

3.8 Net-Benefit Results

Table 14: Net-benefit comparison of treatment arms

Comparison	Global Net Benefit	OS Contribution	PFS Contribution	PFS p-value
B vs A	0.220	0.134	0.086	0.031
C vs A	0.202	0.132	0.069	0.050
C vs B	-0.049	-0.010	-0.039	

The net-benefit analysis supported the same broad interpretation as the win-ratio analysis. Arm B showed a positive global net benefit of 0.220 compared with Arm A, with contributions from OS (0.134) and PFS (0.086; p = 0.031). Arm C showed a positive global net benefit of 0.2015 compared with Arm A, with contributions from OS (0.1323) and PFS (0.0692; p = 0.0499). The comparison between Arms C and B showed no meaningful advantage, with a global net benefit of -0.049.

3.9 Multi-State Model Results

Table 15: Multi-state transition hazard ratios by treatment arm

Treatment Arm	Stable to Progression HR (95% CI)	Progression to Death HR (95% CI)
Arm A (baseline)	1.00	1.00
Arm B	0.56 (0.35–0.88)	1.37 (0.81–2.31)
Arm C	0.59 (0.38–0.92)	1.02 (0.62–1.69)

The multi-state model indicated that Arms B and C reduced the transition hazard from stable disease to progression compared with Arm A. Arm B had a stable-to-progression hazard ratio of 0.56, while Arm C had a stable-to-progression hazard ratio of 0.59. There was no clear evidence that either arm changed the progression-to-death transition when treatment arm alone was included in the multi-state model.

Table 16: Twelve-month transition probabilities from stable disease

Arm	P(Stable at 12 months)	P(Progression at 12 months)	P(Death at 12 months)
A	0.728	0.147	0.125
B	0.838	0.074	0.088
C	0.830	0.092	0.078

Table 17: State occupation probabilities across time by treatment arm

Time	Stable_A	Stable_B	Stable_C	Death_A	Death_B	Death_C
0	1.00	1.00	1.00	0.00	0.00	0.00
2	0.95	0.97	0.97	0.01	0.00	0.00
4	0.90	0.94	0.94	0.02	0.01	0.01
6	0.85	0.92	0.91	0.04	0.03	0.02
8	0.81	0.89	0.88	0.06	0.05	0.04
10	0.77	0.86	0.86	0.09	0.07	0.06
12	0.73	0.84	0.83	0.12	0.09	0.08

Across time, Arms B and C maintained higher probabilities of remaining stable than Arm A. By 12 months, the probability of remaining stable was approximately 0.73 for Arm A, 0.84 for Arm B, and 0.83 for Arm C. The probability of death was also higher for Arm A than for Arms B and C at the same time point.

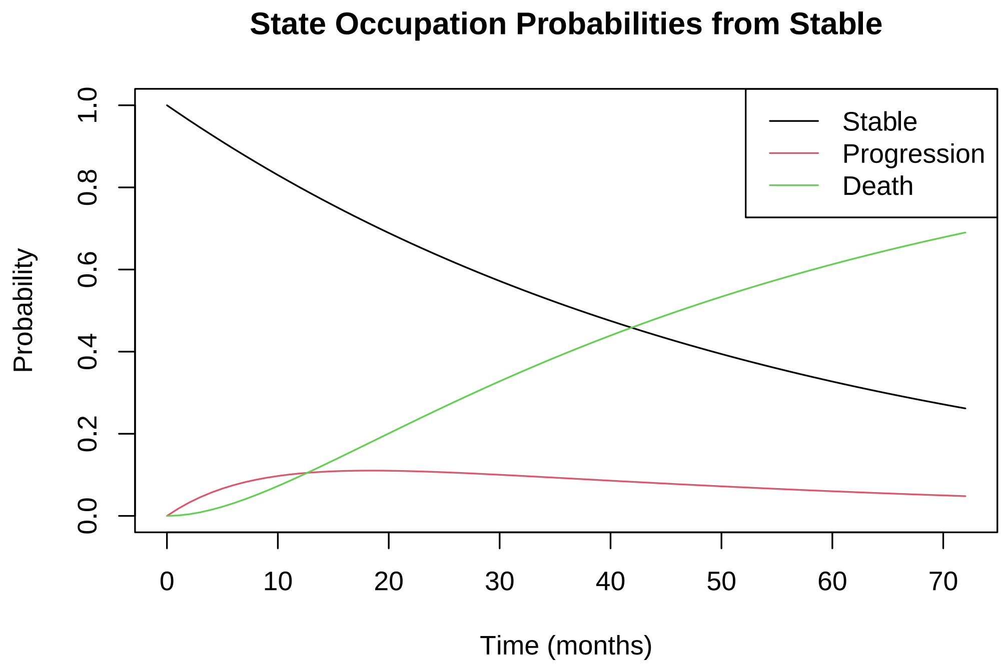

Figure 6: State occupation probabilities from stable disease

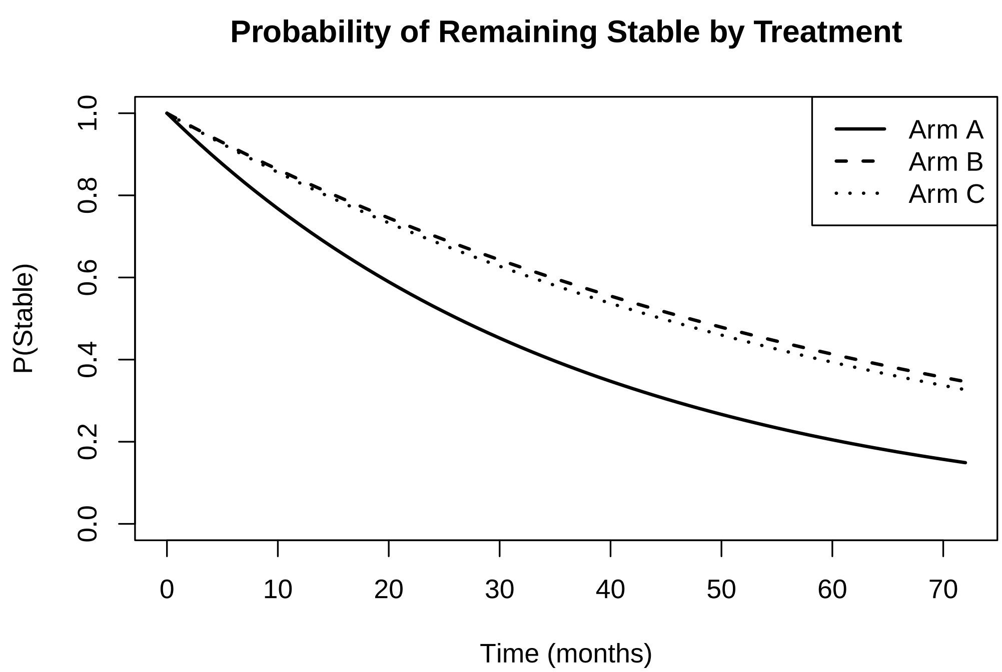

Figure 7: Probability of remaining stable by treatment arm

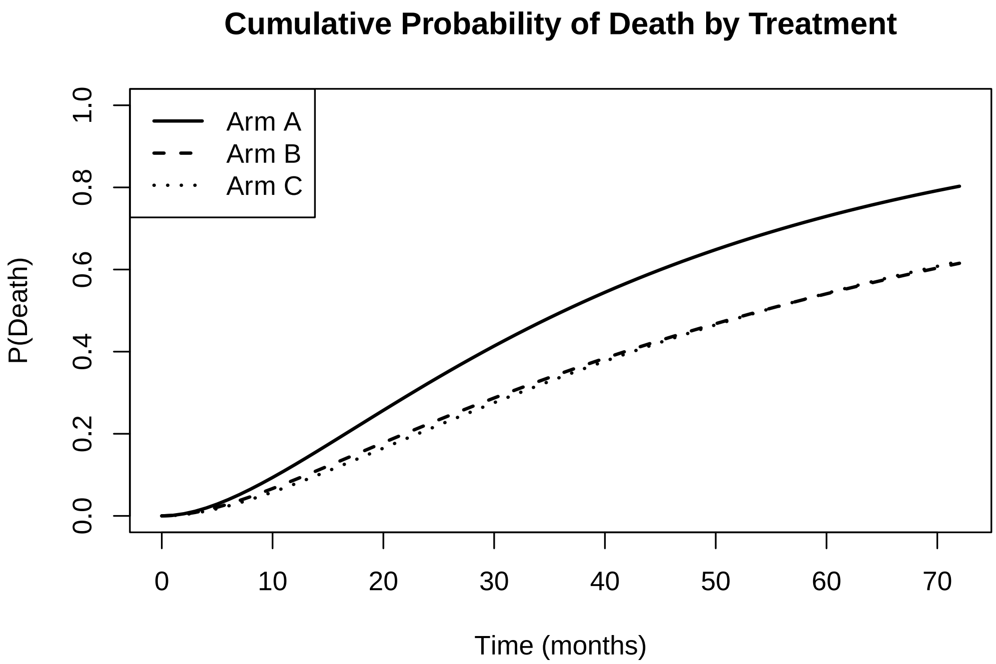

Figure 8: Cumulative probability of death by treatment arm

3.10 Comparison of Methods

Table 18: Comparison of statistical methods for multiple outcomes

Method	Multiple outcomes	Time-to-event	Censoring	Covariates	Output	Limitation
Logistic regression	Composite only	No	No	Yes	Odds ratio	Ignores timing
Kaplan–Meier	Composite only	Yes	Yes	No	Survival curve	No adjustment
Log-rank test	Composite only	Yes	Yes	No	p-value	No effect size
Cox model	Composite only	Yes	Yes	Yes	Hazard ratio	PH assumption
Win ratio	Yes	Yes	Yes	Limited	Win ratio	Needs hierarchy
Net benefit	Yes	Yes	Yes	Limited	Net benefit	Pairwise framework
Multi-state model	Yes	Yes	Yes	Yes	Transition probabilities	Transition assumptions

The comparison shows that logistic regression, Kaplan–Meier analysis, log-rank tests, and Cox models are useful but limited when outcomes are related and sequential. Win-ratio and net-benefit methods better incorporate clinical hierarchy, while multi-state models provide the most complete representation of disease pathways over time.

4 Discussion

4.1 Main Findings

This project compared several statistical strategies for handling multiple outcomes in health data. OS did not show a statistically significant difference across treatment arms, while PFS showed stronger evidence of treatment-arm differences. The composite endpoint analysis was nearly identical to the PFS analysis because no patient died before progression. Win-ratio and net-benefit analyses suggested favourable trends for Arms B and C compared with Arm A. The multi-state model showed that Arms B and C mainly delayed progression from stable disease, rather than clearly changing mortality after progression.

4.2 Interpretation of Treatment Sequencing

The results suggest that treatment sequencing may influence disease progression dynamics. Arm A began with encorafenib plus binimetinib and switched to ipilimumab plus nivolumab after progression. Arm B began with ipilimumab plus nivolumab and switched to encorafenib plus binimetinib after progression. Arm C began with encorafenib plus binimetinib for eight weeks before switching to ipilimumab plus nivolumab until progression.

The multi-state model suggested lower stable-to-progression transition hazards for Arms B and C compared with Arm A. This may indicate that earlier exposure to immunotherapy is associated with delayed progression. However, these findings should be interpreted cautiously because the study had a relatively small sample size and was not powered to definitively test transition-specific treatment sequencing hypotheses.

4.3 Composite Endpoint and PFS Similarity

A key finding was that PFS and the composite endpoint produced almost identical results. This occurred because every progression event was also counted as a composite event, and no patient died before progression. Therefore, in this dataset, the composite endpoint was effectively driven by progression rather than death.

This is important because it shows that the similarity between PFS and composite results was not a coding error. It also highlights a limitation of composite endpoints: if one component dominates the composite, the result may not reflect the full clinical hierarchy of outcomes. In this case, the composite endpoint did not add substantial information beyond PFS.

4.4 Novel Findings

The most clinically interesting finding was the possible effect of treatment sequencing on disease progression. The early Kaplan–Meier curves suggested a change in treatment-arm pattern around the two-month time point, which corresponds to the planned switch in Arm C. This supports the hypothesis that the timing of immunotherapy exposure may influence early disease control.

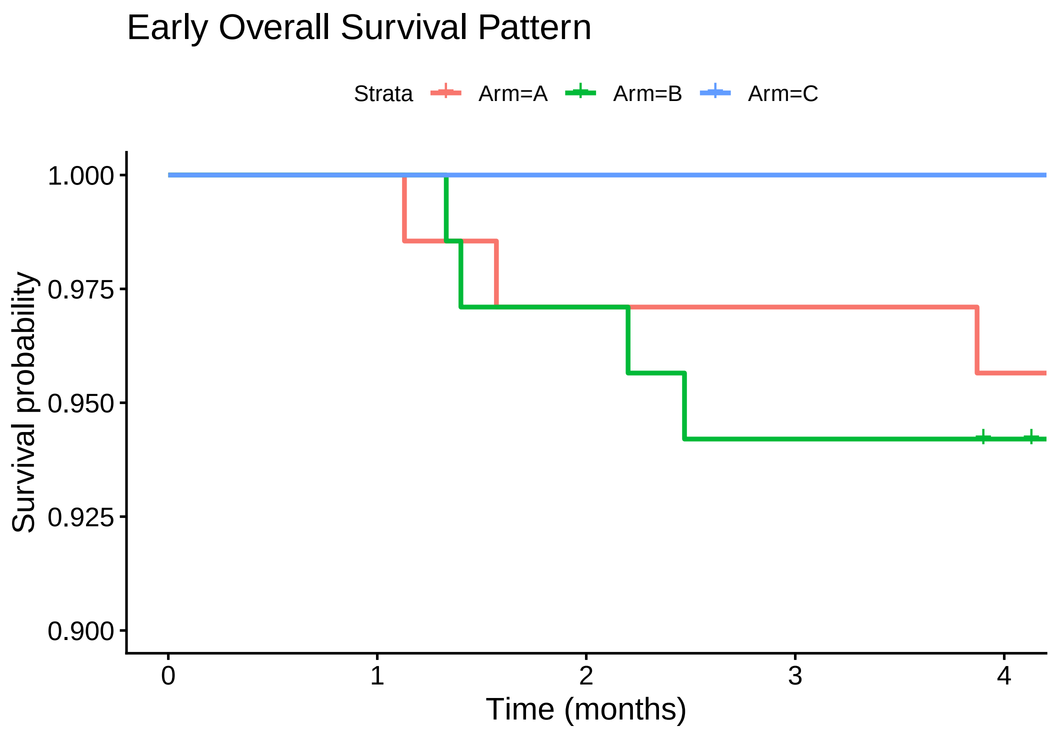

Figure 9: Early overall survival pattern up to four months

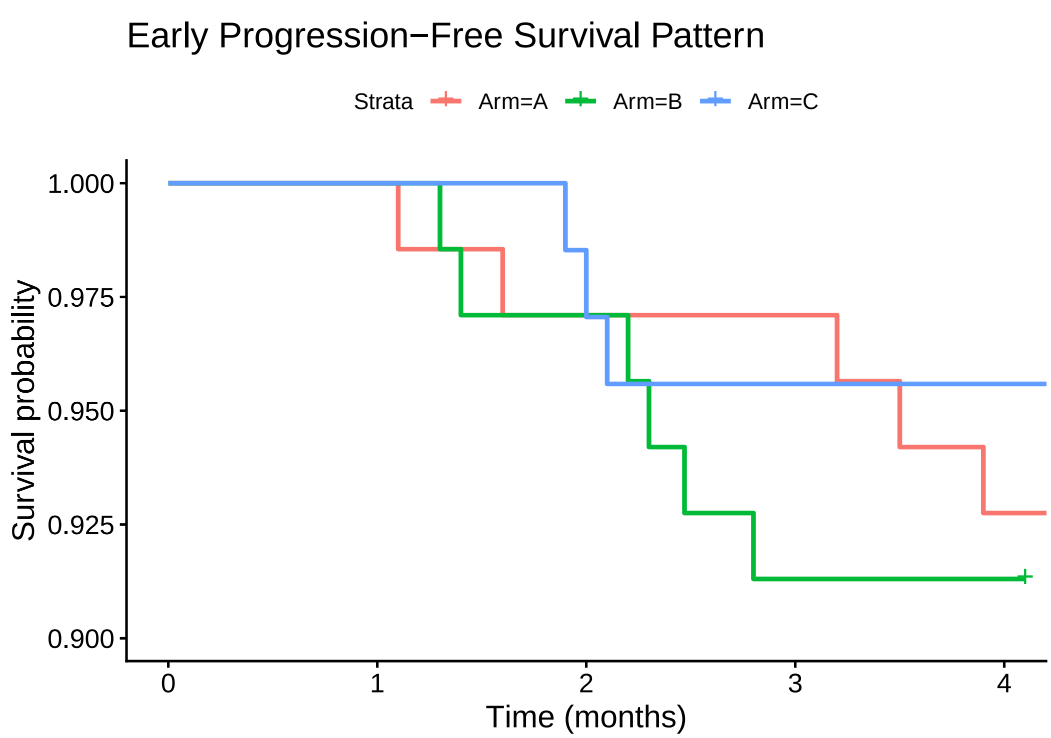

Figure 10: Early progression-free survival pattern up to four months

Inspection of the early curves suggests that the treatment-arm pattern may change around the two-month point. This is clinically important because Arm C switches treatment at approximately eight weeks. Although this finding is exploratory, it suggests that ipilimumab plus nivolumab may contribute to delayed progression when used early in the treatment sequence. This interpretation is consistent with the broader SECOMBIT findings that immunotherapy-first or planned-switch strategies may be favourable for many patients with metastatic BRAF V600-mutated melanoma (20,21).

4.5 Strengths

A strength of this project is that it compared traditional and advanced statistical methods using the same clinical dataset. This allowed direct comparison across methods in terms of censoring, timing, covariate adjustment, clinical hierarchy, and disease pathway interpretation. Another strength is the use of multi-state modelling, which provided clinically interpretable transition probabilities rather than only endpoint-specific hazard ratios.

4.6 Limitations

The main limitation was the relatively small sample size, with approximately 68–69 patients in each arm. This reduced statistical power, especially for pairwise treatment comparisons and transition-specific multi-state modelling. Missingness in the JAK mutation variable reduced the effective sample size for adjusted models. The proportional hazards assumption was also questionable for some endpoints, which limits interpretation of Cox model hazard ratios.

Another limitation was that the dataset contained only one observed transition pathway: stable disease to progression and progression to death. Since no patient died before progression, the multi-state model could not estimate a direct stable-to-death transition. This limits generalisability to datasets where death before progression is observed.

4.7 Future Work

Future research should apply these methods to larger datasets with more complex disease pathways and greater numbers of outcome events. Larger datasets would improve statistical power for transition-specific modelling and would allow direct stable-to-death transitions, competing risks, recurrent events, and time-dependent covariates to be explored.

Future work could also develop dynamic prediction models that estimate an individual patient's probability of remaining stable, progressing, or dying at future time points. This would make multi-state models more directly useful for clinical decision-making. Additional studies are also needed to investigate the potential treatment sequencing effects suggested by the SECOMBIT analysis, particularly the role of early immunotherapy exposure.

5 Conclusion

Traditional survival methods such as Kaplan–Meier analysis, log-rank tests, and Cox models are useful for analysing individual time-to-event outcomes. However, they are limited when outcomes are related, hierarchical, or sequential. Composite endpoints provide a simple combined outcome but may hide important differences between clinically unequal components. Logistic regression is especially limited for this type of data because it ignores event timing and censoring.

The win-ratio and net-benefit methods improved interpretation by incorporating clinical hierarchy and overall treatment advantage. The multi-state model provided the most complete view of disease progression because it described how patients moved from stable disease to progression and death over time.

In this SECOMBIT analysis, Arms B and C appeared to delay progression compared with Arm A, while there was no clear evidence of a difference in post-progression mortality. The findings suggest that advanced methods, particularly multi-state models, can provide more clinically meaningful insight than analysing multiple outcomes separately.

6 Project Link

The code repository for this project is available at: https://github.com/darshu-d/MSc-Research-project-

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